DO NOW — RECALL

Lesson 149 · Client–Server and Peer-to-Peer Network Mo

Name: _____ Class: _____ Date: _____

1. Which device connects several wireless users to an existing school LAN?

2. Which storage type is volatile?

3. Which layer in TCP/IP handles IP routing?

4. What data structure follows FIFO?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A wireless access point.

2. Model answer

RAM.

3. Model answer

The internet layer.

4. Model answer

A queue.

By the end, you can…

• Explain how client–server and peer-to-peer models divide responsibility and resources.

• Compare reliability, administration, cost, security and scalability in context.

• Recognise hybrid systems that use central identity but peer-to-peer data transfer.

Vocabulary — know these terms

client–server — Managed service provided by server to requesting clients.

peer-to-peer — Peers directly share services/resources.

centralisation — Services controlled from a smaller set of authorities.

scalability — How well a system handles growth in load/users.

NEW LEARNING

Read, annotate and explain

Client–server

Many clients request services from a managed central server.

Peer-to-peer

Each participating peer may offer and consume resources.

Select architecture

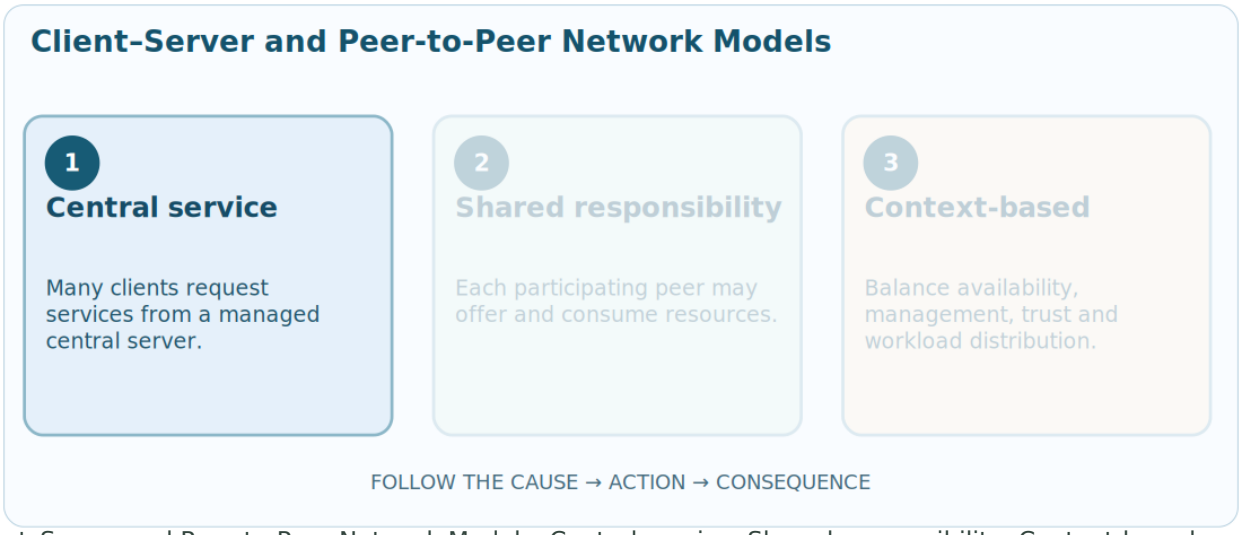
Balance availability, management, trust and workload distribution.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

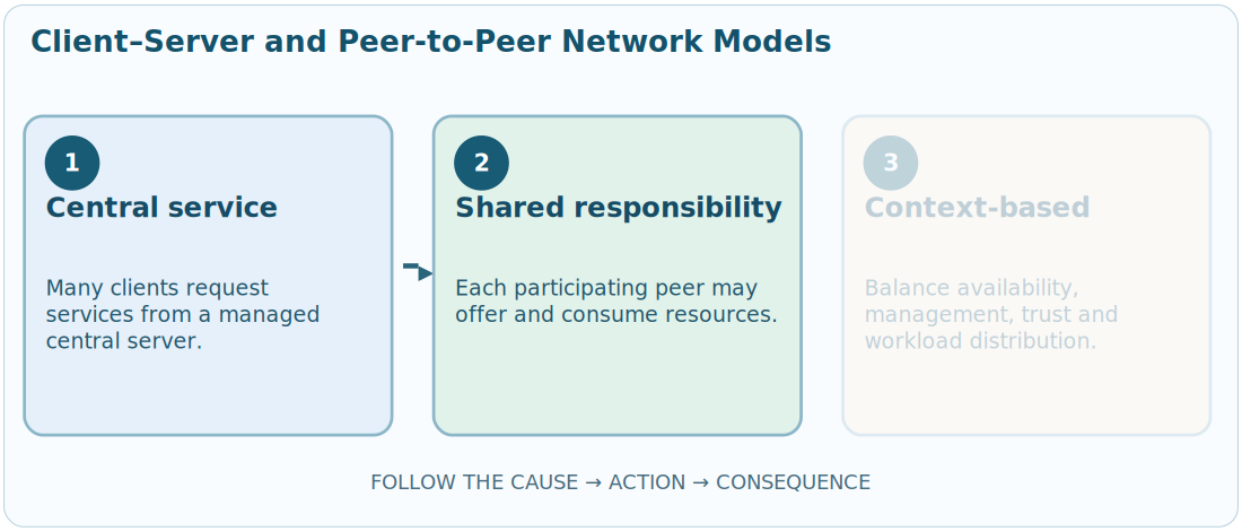
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Client–Server and Peer-to-Peer Network Models: Central service, Shared responsibility, Context-based

New Learning

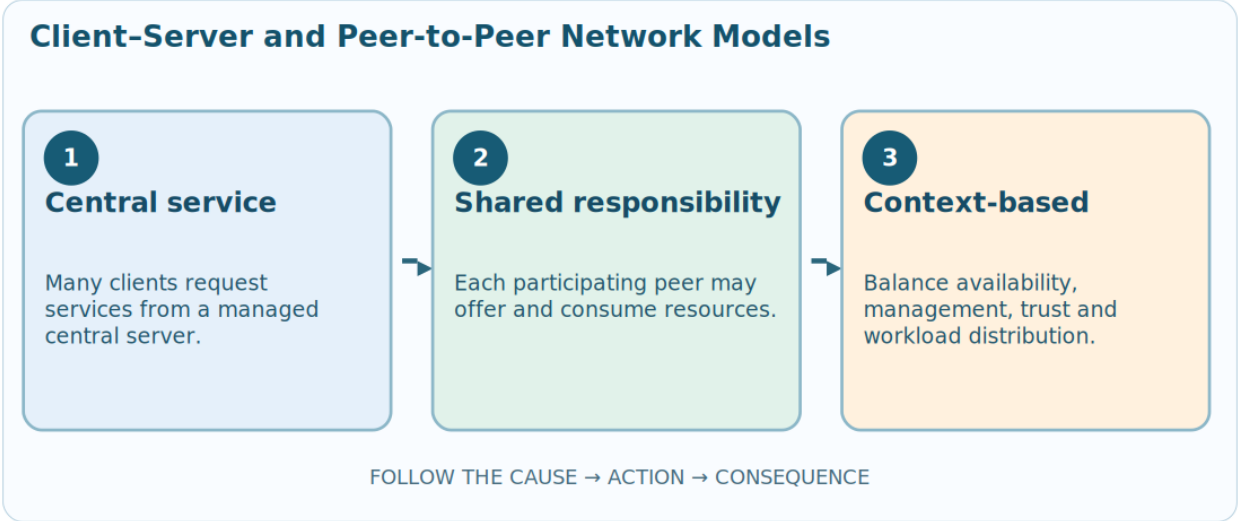


Client–Server and Peer-to-Peer Network Models: Central service, Shared responsibility, Context-based

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Client–Server and Peer-to-Peer Network Models: Central service, Shared responsibility, Context-based

I DO — WORKED EXAMPLE

Follow the worked method

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend a model.

Teachers send requests

Identify the facts and make an initial prediction.

Central marks database

Explain the mechanism or trace the intermediate stage.

Access control + backup

Client–server suits managed central data, controlled permissions, auditing and backups. A server failure can affect many clients, so redundancy and resilient backups matter. Peer-to-peer file sharing is harder to manage consistently.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend

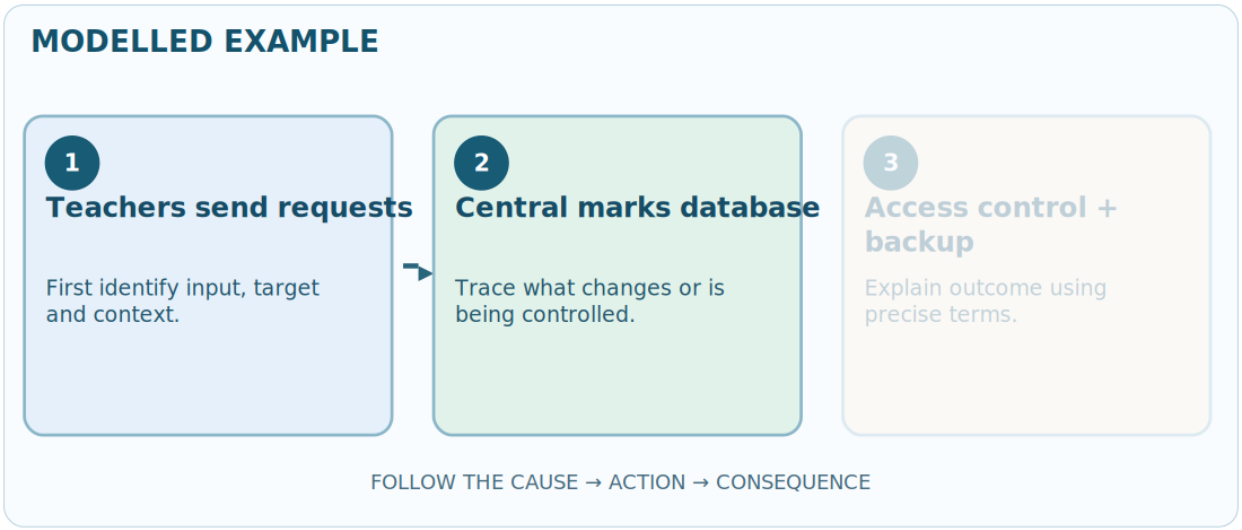
Identify the facts and make an initial prediction.



MODELLED EXAMPLE: Teachers send requests, Central marks database, Access control + backup

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend

Identify the facts and make an initial prediction.



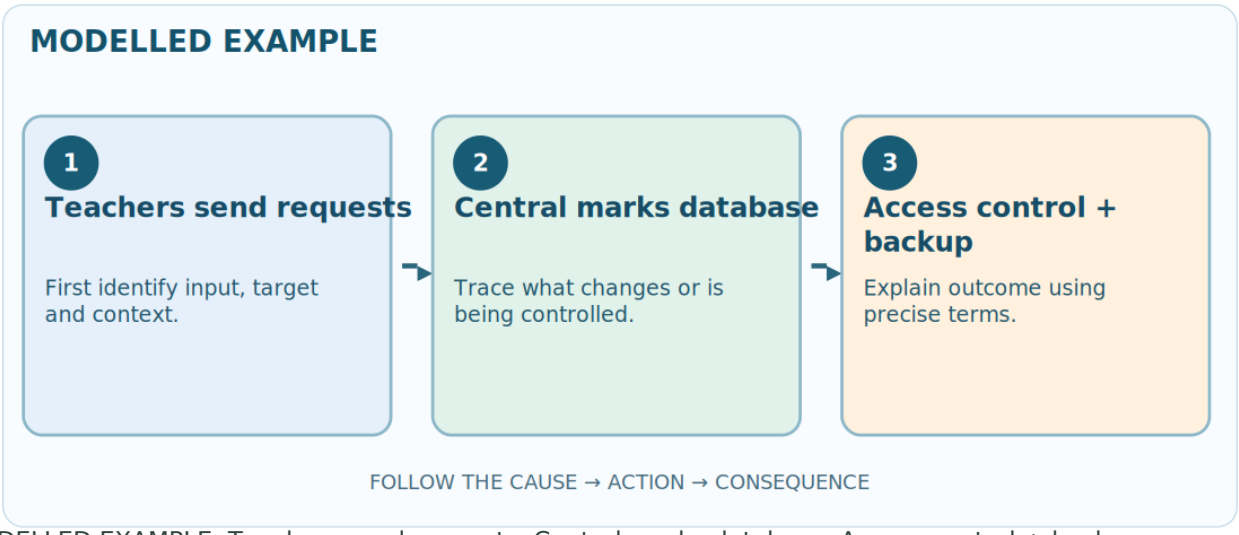
MODELLED EXAMPLE: Teachers send requests, Central marks database, Access control + backup

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend

Identify the facts and make an initial prediction.



MODELLED EXAMPLE: Teachers send requests, Central marks database, Access control + backup

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small group distributes a large open-licensed software image to many volunteers. Is peer-to-peer appropriate

Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Peers hold file blocks, Peers share download load, Verify authentic file

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Peers hold file blocks Identify the facts and make an initial prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define client, server and peer in the context of a network service. [6]

Question 2

Explain two benefits and one limitation of client–server for a school's student database. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare P2P and client–server for distribution of a large video archive, including integrity checks. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A network using client–server can never contain any peer-to-peer activity.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small group distributes a large open-licensed software image to many volunteers. Is peer-to-peer appropriate

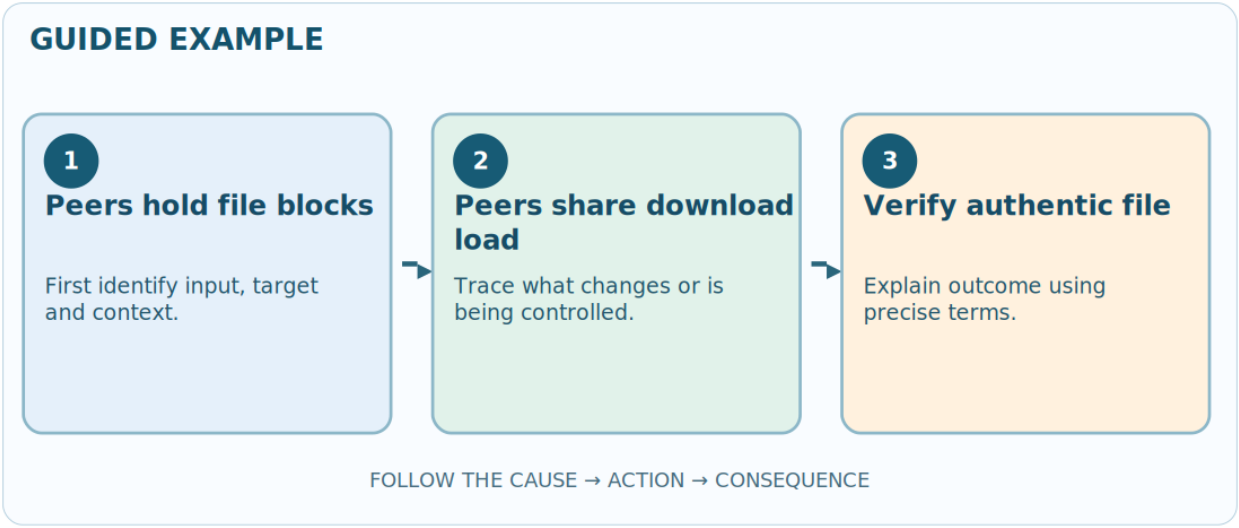
Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Peers hold file blocks, Peers share download load, Verify authentic file

A small group distributes a large open-licensed software image to many volunteers. Is peer-to-peer appropriate

Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Peers hold file blocks, Peers share download load, Verify authentic file

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Peers share download load

Explain the mechanism or trace the intermediate stage.

Verify authentic file

A P2P distribution model can share upload load among peers, reducing demand on one central server. Clients must verify file authenticity with trusted signatures/hashes; peers may be untrusted or offline.

IMPROVE: Explain one correction to your own reasoning.